

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 4: Three-Dimensional Wing Airflow & Induced Drag

Syllabus Source: CAE Oxford (pp. 109-144) | Pass Target: >= 90% AviationExam

1. THREE-DIMENSIONAL AIRFLOW & WINGTIP VORTICES

A real finite wing produces 3D airflow due to spanwise pressure differentials between upper and lower wing surfaces:

- Spanwise Flow: Air on the high-pressure lower surface flows OUTWARD toward wingtips; air on the low-pressure upper surface flows INWARD toward fuselage. At the trailing edge and wingtips, these opposing spanwise flows roll up into trailing WINGTIP VORTICES.
- Induced Downwash (w): Wingtip vortices trail downstream and induce a strong downward vertical velocity component behind the wing called induced downwash. Downwash tilts the local relative airflow vector downwards!
- Effective Angle of Attack: Effective AOA = Geometric AOA - Induced Angle (α_i). Because the relative airflow is tilted downward, the total aerodynamic lift vector tilts REARWARD. The rearward-tilted component of lift is INDUCED DRAG (Vortex Drag)!

2. INDUCED DRAG FORMULA & WING GEOMETRY FACTORS

Induced drag is the inevitable aerodynamic price of generating lift with a finite-span wing:

- Induced Drag Coefficient: $C_{Di} = (C_L^2) / (\pi \times AR \times e)$, where AR = Aspect Ratio, e = Oswald wing efficiency factor (~0.8 to 0.9).
- Variation with Speed: Because C_L is inversely proportional to V^2 in level flight ($C_L = W / (q S)$), Induced Drag is INVERSELY PROPORTIONAL TO SPEED SQUARED: D_i proportional to $1 / V^2$! Induced drag is massive at low speeds (take-off, climb, landing approach) and becomes negligible at high cruising speeds.
- Variation with Mass: Induced drag is directly proportional to the SQUARE OF AIRCRAFT GROSS MASS: D_i proportional to W^2 ! A 20% increase in mass increases induced drag by $(1.20)^2 = 44\%$!

Design Factor	Aerodynamic Influence on Induced Drag	Aeronautical Examples & Trade-offs
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Aspect Ratio (AR = b^2 / S)	HIGH Aspect Ratio dramatically REDUCES induced drag (spreads lift over longer wingspan, reducing vortex strength).	Gliders (AR ~ 25 to 30), high-altitude surveillance (U-2). Limited by wing root bending moments and structural weight.
Wing Planform Taper	Elliptical lift distribution produces minimum induced drag ($e = 1.0$). Tapered wings approximate elliptical distribution.	Supermarine Spitfire (pure elliptical); modern airliners use trapezoidal tapered wings with aerodynamic washout.
Winglets / Sharklets	Vertical wingtip fences diffuse tip vortices, converting vortex energy into forward aerodynamic thrust.	Airbus sharklets / Boeing blended winglets reduce fuel burn by 3% to 5% in cruise.
Wing Washout (Twist)	Geometric wing twist (lower incidence at tip than root).	Ensures wing root stalls FIRST, preserving aileron roll control at high angles of attack.

[!] EXAM TRAP (AVIATIONEXAM) - Wake Turbulence & Heavy Jet Vortex Hazard

- When is Wake Turbulence Greatest?: When the generating aircraft is HEAVY, CLEAN (flaps/gear up), and SLOW! - Rotation / Touchdown Points: Vortices begin at nose-wheel rotation and end when nose-wheel touches down. - Avoidance Rule: Rotate BEFORE the preceding heavy aircraft's rotation point, and land BEYOND its touchdown point!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Aspect Ratio and Induced Drag**

Problem: Two wings have equal wing area $S = 20 \text{ m}^2$. Wing A has span 10 m; Wing B has span 20 m. Which has lower induced drag? Step 1: Compute Aspect Ratio $AR = b^2 / S$. For A: $AR = 10^2 / 20 = 5$. For B: $AR = 20^2 / 20 = 20$. Step 2: Induced drag coefficient C_{Di} is inversely proportional to AR. Wing B has 4 TIMES LESS induced drag than Wing A at the same C_L . Exam Rule: Longer wingspan = higher aspect ratio = lower induced drag.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: How does Induced Drag vary with aircraft airspeed in straight and level flight?**

- [A]: Directly proportional to airspeed
- [B]: Directly proportional to airspeed squared (V^2)
- [C]: Inversely proportional to airspeed SQUARED ($1 / V^2$)
- [D]: It remains constant

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. In level flight, as speed increases, C_L decreases by $1/V^2$. Because induced drag is proportional to C_L^2 , Induced Drag decreases with $1/V^2$, making it dominant at low speeds and tiny at high speeds.

Q2: What is the primary aerodynamic benefit of fitting winglets to modern jet transport aircraft?

- [A]: To increase maximum take-off mass
- [B]: To reduce Induced Drag by impeding spanwise airflow around wingtips and dissipating wingtip vortices
- [C]: To prevent deep stall
- [D]: To improve rudder authority

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Winglets act as vertical aerodynamic barriers that disrupt the high-to-low pressure spanwise bleed around wingtips, reducing vortex downwash and cutting induced drag by 3-5%.

Q3: Under what flight conditions does a transport aircraft generate the strongest, most hazardous wake turbulence?

- [A]: Fast, light, flaps fully extended
- [B]: HEAVY, CLEAN (flaps retracted), and SLOW
- [C]: Cruising at high Mach number
- [D]: In a steep dive

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Wake vortex strength is proportional to aircraft weight and inversely proportional to airspeed and wingspan. A heavy aircraft flying slowly in clean configuration generates maximum vortex circulation.